

The Beta Neutral Cross Sectional Multilayer Perceptron

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Abstract

This document describes the beta neutral cross sectional multilayer perceptron for monthly portfolio construction. The model encodes each security from its characteristics, mixes that encoding with a pooled summary of the whole cross section so that securities inform one another before being scored, and is fitted under a maximum Sharpe ratio regression objective. To this it adds a single risk side operation, namely the removal of the portfolio’s exposure to market beta. We develop the full model from the data representation onward, then present the neutralisation as a linear projection, and finally explain why the projection is constructed without an intercept.

1 Overview and motivation

A portfolio whose weights are driven entirely by characteristics has no reason to be neutral to the market factor. Any residual exposure to market beta is a systematic risk that the selection signal does not intend to take, and on this panel it is unrewarded. The model is therefore designed to retain a characteristic driven selection signal while stripping out the component of the book that merely loads on the market.

2 The cross sectional perceptron and the data pipeline

The neutralisation is placed on a core network and data pipeline that it changes only at the final step. The raw inputs are a panel of monthly characteristics and a list of candidate features, and the target is the excess return of a security over the following month. A characteristic is retained if its coverage is at least ninety percent over the pre test period, which makes the selection causal and algorithmic. For each month every retained characteristic is transformed to its within month percentile rank, missing entries are assigned the median value before centring so that the inputs lie in the interval $[-0.5, 0.5]$, and securities missing more than one third of the characteristics are dropped.

Each security is encoded to an embedding z_i , a pooled context $c = n^{-1} \sum_i z_i$ is formed as the average embedding across the month, and the score is $s_i = \text{head}([z_i, c])$, so that a security’s score depends on the composition of its cross section. The network is fitted under the maximum Sharpe ratio regression loss $(1 - \sum_i w_i r_i)^2$, averaged over a rolling window of at most sixty months whose forward returns are observable at the formation date, with a fifty epoch cold start on the first fit and ten epoch warm starts thereafter, and averaging over three seeds. The seed averaged scores, each normalised to unit gross exposure, form the raw book to which the neutralisation is applied.

3 Beta neutralisation as a projection

Let $w \in \mathbb{R}^n$ denote the seed averaged raw book in a given month and let $b \in \mathbb{R}^n$ denote the vector of market betas of the securities, taken from the field `beta_60m` at the formation date and, in

the implementation, transformed to its centred within month rank so that the projection uses a stable cross sectional ordering rather than raw beta levels. The component of w that lies along b is removed by solving the least squares problem

$$\hat{c} = \arg \min_c \|w - bc\|^2 = \frac{b^\top w}{b^\top b}, \quad (1)$$

and taking the residual as the neutralised book,

$$w^\perp = w - b\hat{c}. \quad (2)$$

The residual satisfies $b^\top w^\perp = 0$ by construction, so the neutralised book has zero exposure to the beta vector. The neutralised weights are then renormalised to unit gross exposure. From the orthogonality condition we can conclude that the book carries no first order sensitivity to the market factor as proxied by b , while every characteristic driven tilt orthogonal to b is preserved.

4 The role of the excluded intercept

A deliberate feature of the construction, and the point on which the design turns, is that the projection regresses w on b alone and does not include an intercept. Had an intercept been included, the projection would also have removed the component of w along the constant vector, which is the net exposure of the book. Excluding the intercept therefore preserves the net tilt of the portfolio while removing only the beta aligned part. The evidence on this panel records that removing the net exposure as well, that is including the intercept, costs approximately half a Sharpe point, whereas the residual market beta that the projection removes is unrewarded risk. It is therefore safe to conclude that the net exposure carries reward on this panel and should be retained, while the beta exposure should not.

5 Temporal integrity and evaluation

The neutralisation introduces no lookahead, since the beta used at each formation date is the value known at that date and the projection is a contemporaneous cross sectional operation. The book is evaluated by the annualised Sharpe ratio of its monthly returns, both unscaled and scaled to a ten percent annual volatility target through a trailing and lagged leverage overlay that changes the time path of leverage but not the underlying weights.

On the evaluation window the model attains a Sharpe ratio of 2.541 in the volatility scaled form and 2.614 in the unscaled form. An otherwise identical model without the neutralisation attains 2.484 on this panel, so the improvement is modest but consistent with the interpretation that removing an unrewarded systematic exposure raises the Sharpe ratio of an unchanged selection signal.

6 Concluding remarks

The beta neutral model isolates a clean risk side intervention. The selection machinery is left untouched, and a single linear projection removes an exposure that the panel does not reward while preserving the net tilt that it does. The result illustrates a theme that recurs on this panel, namely that aggregate risk side adjustments applied to the whole book tend to help, whereas finer per security risk modelling tends not to. Within the limitations of the present analysis, the neutralisation is a low cost and interpretable refinement of a characteristic driven portfolio.